

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Developed and optimized **deep learning models** for neural activity decoding, achieving **15% improvement** in signal interpretation accuracy by implementing novel **generative AI** architectures.
- Engineered and maintained **machine learning pipelines** for distributed environments, reducing model training time by **20%** through efficient data preprocessing and **GPU utilization**.
- Collaborated with research scientists to advance **representation learning** initiatives, contributing to **3 research publications** in peer-reviewed journals.
- Implemented **Python-based automation scripts** for data ingestion and feature engineering from large, unstructured datasets, accelerating downstream analytics by **30%**.
- Maintained high standards of **software engineering** and **code quality** across AI projects, ensuring reproducibility and reducing integration bugs by **25%** through rigorous testing.
- Identified and resolved technical bottlenecks in **AI infrastructure**, enhancing model deployment efficiency by **18%** across diverse computing platforms.

SKILLS

Machine Learning & Deep Learning – PyTorch, TensorFlow, Generative AI, Representation Learning, Neural Networks, Model Optimization, Supervised Learning, Unsupervised Learning

Programming & Scripting – Python, SQL, R

Data Handling & Analysis – Large Datasets, Unstructured Data, Data Warehousing, ETL, Pandas, NumPy, Scikit-learn

AI Infrastructure & Tools – Hugging Face Transformers, Diffusers, Accelerator, GPU Computing, Docker, CI/CD, Jenkins, Git

Cloud & Systems – AWS (EC2, S3, Lambda, CloudWatch), Linux

Business Intelligence – Power BI, DAX, Tableau, Data Modelling

Software Engineering – REST APIs, Java Spring Boot, Microservices, Code Quality, Reproducibility

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Neural Activity Decoding Engine (Portfolio Project)

- Developed a scalable **deep learning model** using **PyTorch** for decoding complex neural signals, achieving **92% accuracy** in identifying cognitive states.

- Engineered a robust **machine learning pipeline** integrating **Hugging Face Transformers** and **GPU systems** for rapid experimentation and training on terabyte-scale datasets.
- Implemented **Python scripts** for data preprocessing and feature extraction from **unstructured neural data** (EEG/fMRI concepts), improving model performance by **18%** and reducing processing latency.

Financial Transaction Anomaly Detection (HCL Technologies)

- Built a Python-based anomaly detection system for a **payments operations dashboard**, identifying suspicious transactions with **95% precision** and reducing manual review efforts by **40%**.
- Developed **SQL queries** against PostgreSQL and Oracle databases to extract and validate transaction data for model training and operational reporting.
- Integrated the anomaly detection service via **REST APIs** into the existing financial services platform, improving real-time fraud monitoring capabilities.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)